Joseph H. Rudoler

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Education

University of Pennsylvania

2023 – **Ph.D., Statistics and Data Science**, *The Wharton School*
Advisors: Giles Hooker and Konrad Körding

2021 – 2023 **M.S.E., Data Science**, *School of Engineering and Applied Science*
Thesis title: Transfer learning to improve memory state classification from EEG.

2016 – 2020 **B.A., Physics and Astronomy (*Magna Cum Laude*)**, *School of Arts and Sciences*
Minors: English, Music

Employment

Computational Memory Lab (PI: Michael Kahana), U. of Penn.

2021 – 2023 **Data and Programming Specialist**
Staff data scientist, managing data pipelines, software development

2020 – 2021 **Clinical Research Specialist**
Coordinating multi-site project, collecting iEEG data from epilepsy patients, maintaining clinical imaging and electrode localization pipeline

Radian Group, Inc.

2019 **Pricing Intern**
Developing pricing models for mortgage insurance premiums, efficient data processing, dashboards / data visualizations

Research

In Preparation

1. **Rudoler**, **J. H.**, Asemota, A., & Hooker, G. (2026). *Targeted learning for inference on data fairness* [Manuscript in preparation].
2. **Rudoler**, **J. H.**, DeHaan, R., Kording, K. P., & Dobriban, E. (2026). *Do we decode better than chance? Efficient hypothesis testing for neural decoders* [Manuscript in preparation].

Preprints

1. **Rudoler**, **J. H.**, Tan, K., Tessler, B., Kong, T., & Boix Adserà, E. (2026, September 28). *Improving scalable oversight with co-trained monitors*. arXiv: 2609.36049 [cs.LG]. https://doi.org/10.48550/arXiv.2609.36049

Peer-Reviewed Papers

1. **Rudoler**, **J. H.**, Saxe, A. M., & Kording, K. (2026). Weight space gradient descent induces approximate activity descent, making learning rules testable. *NeurIPS 2026 Workshop on Symmetry and Geometry in Neural Representations (NeurReps), Extended Abstract Track*. [Accepted].
2. **Rudoler**, **J. H.**, Tan, K., Hooker, G., & Kording, K. P. (2026). Estimating Implicit Regularization in Deep Learning. *NeurIPS 2026*. https://doi.org/10.48550/arXiv.2605.05436 [Accepted].

3. Dougherty, M. R., Chang, W., **Rudoler**, **J. H.**, Katerman, B. S., Halpern, D. J., Bruska, J. P., Diamond, N. B., & Kahana, M. J. (2024). Neural correlates of memory in an immersive spatiotemporal context. *Journal of Experimental Psychology: Learning, Memory, and Cognition*. https://doi.org/10.1037/xlm0001341

4. Kahana, M. J., Lohnas, L. J., Healey, M. K., Aka, A., Broitman, A. W., Crutchley, P., Crutchley, E., Alm, K. H., Katerman, B. S., Miller, N. E., Kuhn, J. R., Li, Y., Long, N. M., Miller, J., Paron, M. D., Pazdera, J. K., Pedisich, I., **Rudoler**, **J. H.**, & Weidemann, C. T. (2024). The Penn Electrophysiology of Encoding and Retrieval Study. *Journal of Experimental Psychology: Learning, Memory, and Cognition*. https://doi.org/10.1037/xlm0001319

5. **Rudoler**, **J. H.**, Bruska, J. P., Chang, W., Dougherty, M. R., Katerman, B. S., Halpern, D. J., Diamond, N. B., & Kahana, M. J. (2024). Decoding EEG for optimizing naturalistic memory. *Journal of Neuroscience Methods*, *410*, 110220. https://doi.org/10.1016/j.jneumeth.2024.110220

6. **Rudoler**, **J. H.**, Herweg, N. A., & Kahana, M. J. (2023). Hippocampal Theta and Episodic Memory. *Journal of Neuroscience*, *43*(4), 613–620. https://doi.org/10.1523/JNEUROSCI.1045-22.2022

Presentations

1. **Rudoler**, **J. H.** *Do We Decode Better than Chance? Efficient Hypothesis Testing for Neural Decoders*. Cognitive Computational Neuroscience, Amsterdam, Netherlands, 2025. (Poster).

2. **Rudoler**, **J. H.** *Decoding and Optimizing Episodic Memory*. Cognitive Neuroscience Society, San Francisco, California, USA, 2022. (Poster).

3. **Rudoler**, **J. H.** *Decoding and Optimizing Episodic Memory*. MathPsych, Toronto, Canada, 2022. (Poster).

4. **Rudoler**, **J. H.** *Oscillatory and Fractal Biomarkers of Human Memory*. Computational and Systems Neuroscience (COSYNE), Lisbon, Portugal, 2022. (Poster).

Datasets

1. Kahana, M. J., **Rudoler**, **J. H.**, Lohnas, L. J., Healey, K., Aka, A., Broitman, A., Crutchley, E., Crutchley, P., Alm, K. H., Katerman, B. S., Miller, N. E., Kuhn, J. R., Li, Y., Long, N. M., Miller, J., Paron, M. D., Pazdera, J. K., Pedisich, I., & Weidemann, C. T. (2023). *Penn Electrophysiology of Encoding and Retrieval Study (PEERS)*. OpenNeuro. https://doi.org/10.18112/openneuro.ds004395.v2.0.0

2. **Rudoler**, **J. H.**, Dougherty, M. R., Katerman, B. S., Bruska, J. P., Chang, W., Halpern, D. J., Diamond, N. B., & Kahana, M. J. (2023). *"Spatial memory and non-invasive closed-loop stimulus timing"*. OpenNeuro. https://doi.org/10.18112/openneuro.ds004706.v1.0.0

Teaching

2026	**STAT4730/7730, Data Science with ChatGPT** *Teaching Assistant* Undergraduate and MBA course on data analysis with LLMs.
2025 –	**STAT4010, Sports Analytics Capstone** *Teaching Assistant* Undergraduate capstone in statistical analysis of sports data.

Teaching (continued)

2025 **US-Israel Academic Bridge Fellowship: Data Science Track**
Instructor
Designed and taught a 5-week summer course in statistics and data science for Israeli undergraduate exchange students. New curriculum emphasizes sampling variability and computational approaches.
Course site: https://jrudoler-teaching.github.io/understanding-uncertainty/

2023 – **STAT6130, Regression Analysis for Business**
Teaching Assistant
MBA core curriculum course on statistical foundations and regression

2023 – 2025 **Wharton Moneyball Academy**
Lecturer
Summer course for high school students in sports analytics and data science

2021 **Estes Summer Workshop in Model-based Cognitive Electrophysiology**
Lecturer
2-week summer course, mostly for doctoral and postdoctoral researchers, in analyzing EEG data to study behavior. Wrote new lecture materials focusing on analyzing periodic and aperiodic components of EEG.

2019 – 2020 **CIS105, Computational Data Exploration**
Tutor (individual)

2017 – 2019 **PHYS150/151, Physics I (Mechanics) and II (Electricity & Magnetism)**
Tutor (open office hours)

Awards, Funding, and Certifications

Awards and Honors

2026 **Donald S. Murray Prize for Excellence in Teaching**
Statistics and Data Science Department, The Wharton School.

2023 **Honorable Mention for Best Thesis**, M.S.E. in Data Science.

2022 **Finalist, NFL Big Data Bowl**, NFL Football Operations
$15,000 team prize for "Optimal Run Path for Kick Returners."

2017 – 2020 **Dean's List**, U. of Penn.

Funding and Fellowships

2026 **Wharton AI & Analytics Initiative Funding**
Teaching statistics and data science from first principles: a data-oriented, AI-enabled course (with Adi Wyner).

2025 – 2026 **Tel Aviv University Research Fellowship for Outstanding International Students**
Visiting research in the Blavatnik School of Computer Science and AI.

2019 **Jeffrey Greenberg Undergraduate Research Fellowship**, U. of Penn.

Certifications

2025 **Certificate in College and University Teaching**
Awarded by Penn's Center for Excellence in Teaching, Learning, and Innovation.